1. Let m be the Lebesgue measure on $[0,1]$ and $I_A(x)$ denote the indicator function of A. Define a sequence of random variables on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], P = m)$: For $n \geq 1$, $X_n(\omega) = 1 + nI_{[0,\frac{1}{n}]}(\omega)$ and $X(\omega) = 1$ for $\omega \in \Omega$.

(10+10+10+15 = 45 points)

   1. (i) Show $X_n \to X$ almost surely (P).

1. (ii) Does $X_n \to X$ in probability ?

1. (iii) Is the class $\{X_n\}$ uniformly integrable (U. I)?

1. (iv) Show $X_n \to X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 1$, but this convergence does not hold for $1 \leq p < \infty$.

(2) Let X, Y, Z be random variables on some probability space and $1 < p, q < \infty$.
(10+15 = 25 points)

2. (a) Let $Y \in L^p, Z \in L^q$ where $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$, then show $YZ \in L^r$.

2. (b) Show that $E\left(e^{X^2}\right) \geq e^{E(X^2)}$

(10+15 = 25 points)

(3) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ-finite measures on it. Define the set function: $\mu_3 \equiv 3\mu_1 + 2\mu_2$ (i.e., $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A)$ for all $A \in \mathcal{F}$). Show: (10+(5+5)+10=30 points)

3. (a) μ_3 is a measure.

3. (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$. Also, express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$and $\frac{d\mu_2}{d\nu}$ (proof required).

3. (c) Let $\mu = \mu_3$ be defined as above where $\mu_1 = Binomial(10, 0.5)$, $\mu_2 = Normal(1, 1)$. Find the Lebesgue decomposition of μ with respect to $\nu = m =$ the Lebesgue measure on $\mathbb{R}$, and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^{1} p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1$, $n \in \mathbb{N}$	$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b])/(b-a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$, $a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha,\beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x) m(dx)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}$, $\sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2/2\sigma^2}}{x} I_{(0,\infty)}(x) m(dx)$